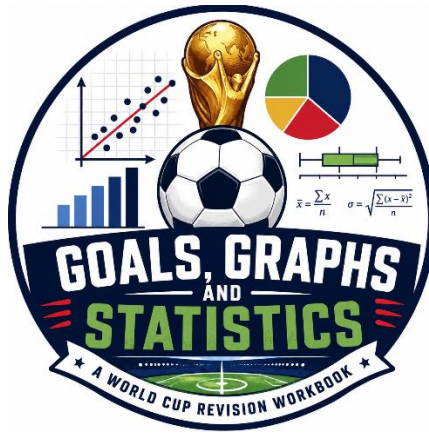


National 5 Applications of Maths

The World Cup Stats Challenge



Question 1: Goals Scored Per Match

The number of goals scored by a Portugal in 10 matches was:

2, 3, 1, 4, 2, 3, 5, 2, 4, 4

- a) Calculate the mean.
- b) Calculate the standard deviation.
- c) Compare your answer to Scotland whose standard deviation is 0.8.

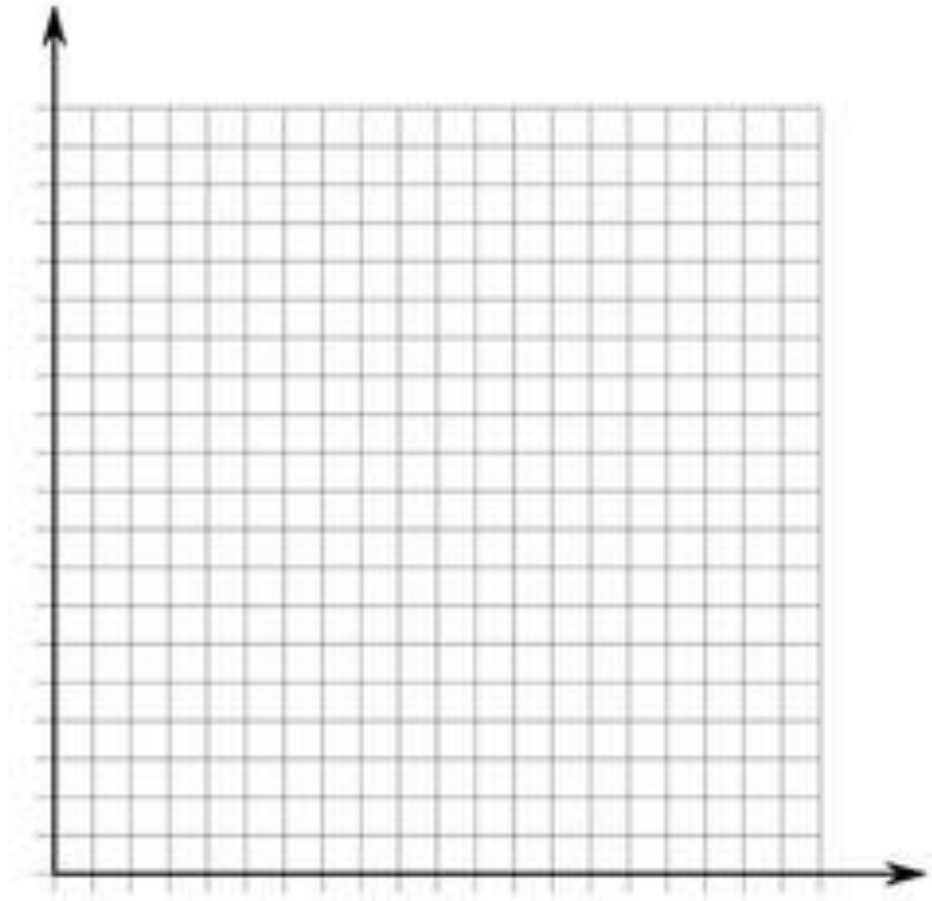
$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Question 2: Goals vs Shots on Target

The table shows data for eight World Cup teams.

Team	Shots on Target	Goals Scored
Argentina	28	12
France	25	10
England	22	8
Brazil	30	13
Morocco	16	5
Japan	18	6
Croatia	14	4
Scotland	20	7

- Draw a scatter graph.
- Describe the correlation.
- Draw a line of best fit.
- Estimate the number of goals a team might score if they have 24 shots on target.



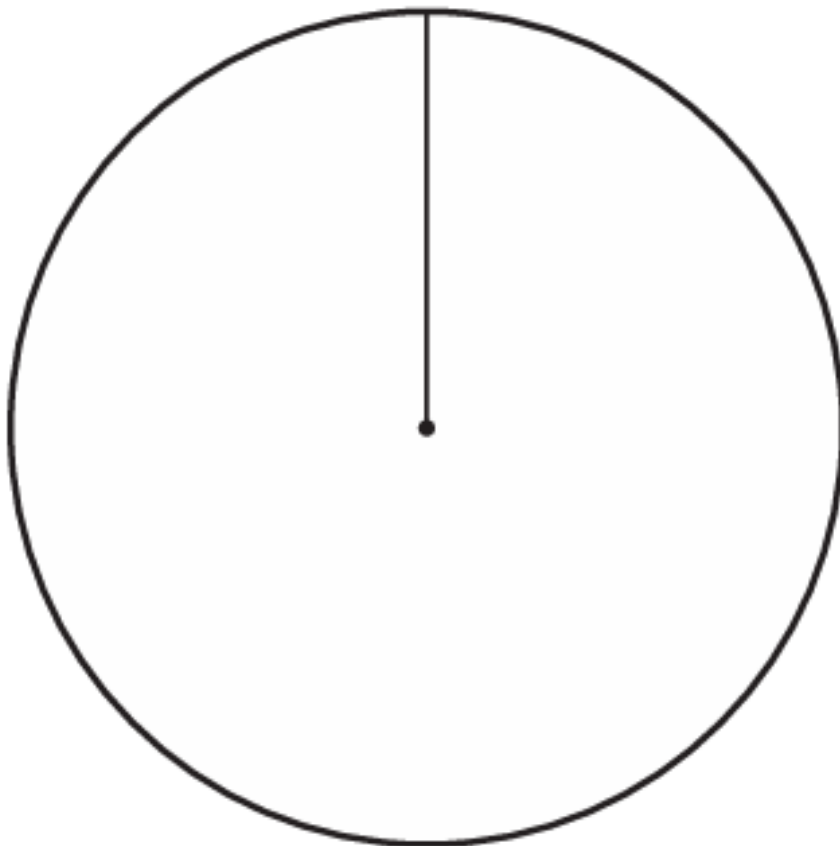
Question 3: World Cup Goals by Method

The last winner of the World Cup was Argentina.

The types of goals they scored are below.

Method	Number of Goals	Angle
Open Play	8	
Set Pieces	3	
Penalties	4	

- Calculate the angle for each sector.
- Draw a pie chart.
- Explain one advantage and one disadvantage of using a pie chart for this data.



Question 4: Ages of Teams

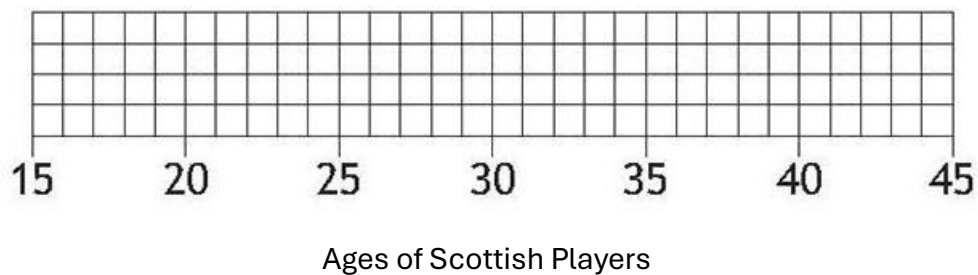
The ages of the Scotland team playing in their opening game against Haiti was

20, 24, 26, 29, 29, 29, 30, 30, 31, 32, 34

a) Find the five-figure summary of the teams ages:

- Minimum
- Lower Quartile (Q1)
- Median
- Upper Quartile (Q3)
- Maximum

b) Draw a box plot.



c) Calculate the interquartile range (IQR).

d) The Interquartile range (IQR) of the Haiti team was 4. Make a comment comparing the spread of ages in the Scotland and Haiti teams.

Question 5: 2022 World Cup Tolerance

1. The ages of the defenders of the Scotland team are below

Scott McKenna	29
Jack Hendry	31
John Souttar	29
Dominic Hyam	30
Grant Hanley	34
Kieran Tierney	29
Andrew Robertson	32
Aaron Hickey	24
Nathan Patterson	24
Anthony Ralston	27

The coach has said the defenders ages are 27 ± 3 years.
State which players do not meet this tolerance.

2. The attendance at the 2022 World Cup Final was 88,966 spectators.

The organisers set a target attendance of 90,000 spectators with a tolerance of $\pm 2\%$.

- a) Calculate the minimum and maximum number of spectators that were expected

- b) Was the organisers target met?

Question 6: 2026 World Cup Stadiums rounding to significant figures and percentages

1. The capacity of each stadium at the 2026 World Cup is shown in the table below.

a) Round each stadium to 1 significant figure and to 2 significant figures:

City	Stadium	Capacity	Capacity (1sf)	Capacity (2sf)
Mexico City	Estadio Azteca	80,824		
New Jersey	Metlife Stadium	80,663		
Dallas	AT&T Stadium	70,649		
Los Angeles	SoFi Stadium	70,492		
Kansas City	Arrowhead Stadium	69,045		
San Francisco	Levi's Stadium	68,827		
Houston	NRG Stadium	68,777		
Philadelphia	Lincoln Financial Field	68,324		
Atlanta	Mercedes-Benz Stadium	68,239		
Seattle	Lumen Field	66,925		
Miami	Hard Rock Stadium	64,478		
Boston	Gillette Stadium	64,146		
Vancouver	BC Stadium	52,497		
Monterrey	Estadio BBVA	51,243		
Guadalajara	Estadio Akron	45,664		
Toronto	BMO Field	43,036		

b) What percentage of stadiums round to 70,000 to 1 significant figure?

c) What percentage of stadiums round to 64,000 to 2 significant figures?

Question 7: Scottish Fans Appreciation and Depreciation

1. JD Sports sold 184 Scotland tops in March 2026; they expected sales to increase by 35.5% over the next three months. How many Scotland tops should they expect to sell in June 2026.
Give your answer correct to 2 significant figures
2. A total of 1540 Scotland fans planned to travel to the World Cup.
Following Scotland's qualification, bookings increased by 62.5%.
Following a change in travel regulations 8.4% of those fans cancelled their trips.
How many fans eventually travelled? Give your answer correct to 3 significant figures
3. A Scotland supporters' group had 2,500 wristbands to sell at the FIFA World Cup.
At the end of each match day, the number of wristbands remaining decreased by 15%.
After 4 match days, how many wristbands remained?
Give your answer correct to 3 significant figures.
4. A shop sold 500 Scotland scarves in January.
Sales increased by 22% in February and increased by a further 22% in March.
In April, sales decreased by 6%.
How many scarves were sold in April?
Give your answer correct to 3 significant figures.

Question 8: Family Trip to Boston

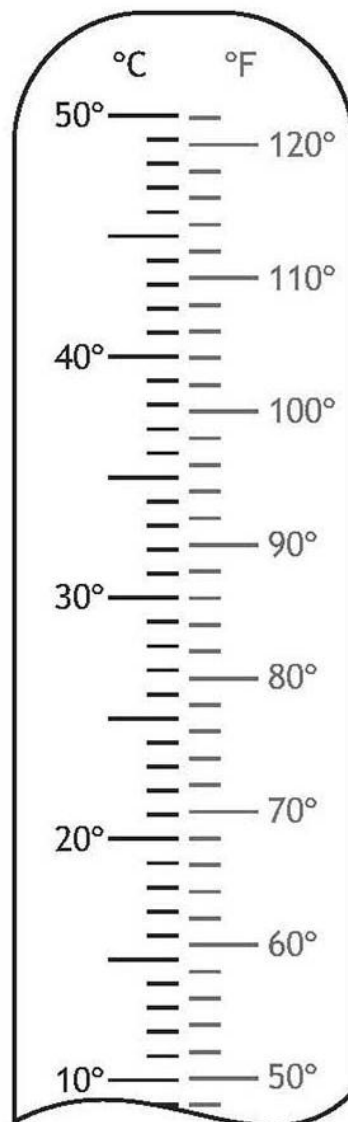
1. A family consisting of two adults and two children who are 4 and 7, are travelling to Boston to watch Scotland play in the world cup. They are looking to book a hotel room close to the venue.

Hilton BOSTON PARK PLAZA		
2 Double Beds Guestroom View room details 	2 Queen Beds Superior Guestroom View room details 	2 Double Beds Deluxe Guestroom View room details 
£420 per night	£465 per night	£515 per night
Special Offer: Save 18% on room bookings over four days		
Extras:	Wi-Fi Rate: £6 per day	
	Valet Parking: £70 per day	
	Breakfast: £25 per day per person Children under 12 pay 50% of this price	

- The family plan to stay from the 10th June until the 22nd June for ten nights
- The want to book a 2 Double Beds Deluxe Guestroom
- They want to purchase Wi-Fi for each day of their stay
- They plan to have breakfast at the hotel for 8 of the nights they are in Boston
- The family have budgeted £5100 for their stay.

Do they family have enough money to choose this hotel?

2. The average temperature in Boston for June is 81°F . When the family leave Scotland the temperature is 16°C . Mark both these temperatures on the scale and state the average temperature in Boston in $^{\circ}\text{C}$.



3. The flight leaves Edinburgh at 16:25 and takes 6 hours and 45 minutes to fly to Boston. The time in Boston is 5 hours behind the time in the UK. What time will it be in Boston when the family arrive?

4. The family have £3250 which they change into dollars. They go to Marks and Spencer's and use the exchange rate given in the table below.
- a) Fill in the table below how many dollars they will receive.

Currency

 **USD** US dollar

Amount

£

2500

=

\$

You'll pay £2500.00

Standard Rate

£1 = \$1.3092

- b) During their 10-day stay in Boston, the family spends \$232.18 each day. They then change their money back into pounds for returning home. How much do they have left at the end of their trip?

National 5 Applications of Maths - The World Cup Stats Challenge **Answers**

Question 1: Goals Scored Per Match

d) $\bar{x} = \frac{30}{10} = 3$

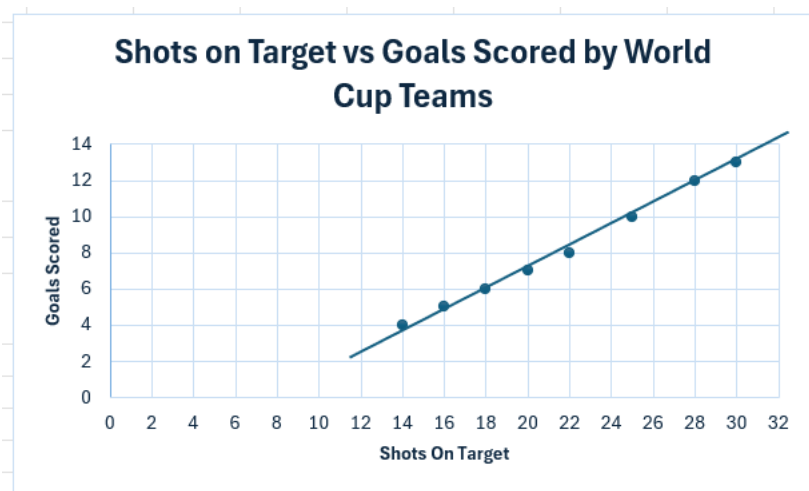
b) $s = \sqrt{\frac{14}{9}} \approx 1.25$

c) The goals scored by Scotland are less varied than the goals scored by Portugal.

The goals scored by Portugal are more varied than the goals scored by Scotland.

Question 2: Goals vs Shots on Target

- a) See graph b) Strong Positive Correlation c) See Graph d) 10 goals



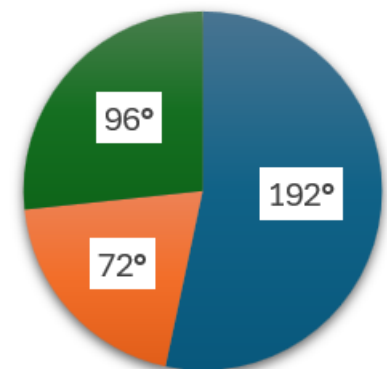
Question 3: World Cup Goals by Method

a)

b)

Method	Number of Goals	Angle
Open Play	8	192°
Set Pieces	3	72°
Penalties	4	96°

Number of Goals by Method



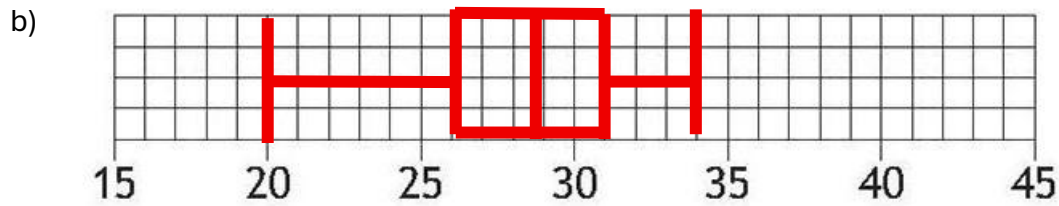
■ Open Play ■ Set Pieces ■ Penalties

c) Positive – Highlights smallest/biggest numbers quickly

Negative – Categories can be similar in size

Question 4: Ages of Teams

a) Minimum – 20 Lower Quartile (Q1) - 26 Median – 29 Upper Quartile (Q3) – 31 Maximum - 34



Ages of Scottish Players

c) $31 - 26 = 5$

d) The ages of Scottish players are more varied than the ages of Haiti players.

Question 5: 2022 World Cup Tolerance

1. a) $27 + 3 = 30$ $27 - 3 = 24$

Jack Hendry Grant Hanley Andrew Robertson

2. a) $2\% = 1800$ $90000 + 1800 = 91800$ $90000 - 1800 = 88200$

b) The organisers met their target as 8896688200

Question 6: 2026 World Cup Stadiums rounding to significant figures and percentages

1.The capacity of each stadium at the 2026 World Cup is shown in the table below.

d) Round each stadium to 1 significant figure and to 2 significant figures:

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Dallas	AT&T Stadium	70,649	70,000	71,000
Los Angeles	SoFi Stadium	70,492	70,000	70,000
Kansas City	Arrowhead Stadium	69,045	70,000	69,000
San Francisco	Levi's Stadium	68,827	70,000	69,000
Houston	NRG Stadium	68,777	70,000	69,000
Philadelphia	Lincoln Financial Field	68,324	70,000	68,000
Atlanta	Mercedes-Benz Stadium	68,239	70,000	68,000
Seattle	Lumen Field	66,925	70,000	67,000
Miami	Hard Rock Stadium	64,478	60,000	64,000
Boston	Gillette Stadium	64,146	60,000	64,000
Vancouver	BC Stadium	52,497	50,000	52,000
Monterrey	Estadio BBVA	51,243	50,000	51,000
Guadalajara	Estadio Akron	45,664	50,000	46,000
Toronto	BMO Field	43,036	40,000	43,000

e) What percentage of stadiums round to 70,000 to 1 significant figure?

50%

f) What percentage of stadiums round to 64,000 to 2 significant figures?

12.5%

Question 7: Prices of Scotland Tops

1. $184 \times 1.355^3 = 457.75 > 480$ tops
2. $1540 \times 1.625 = 2502.5 \times 0.916 = 2292.29 > 2290$ fans
3. $2500 \times 0.85^4 = 1305.01 > 1310$
4. $500 \times 1.22^2 = 744.2 \times 0.94 = 699.548 > 700$

Question 8: Family Trip to Boston

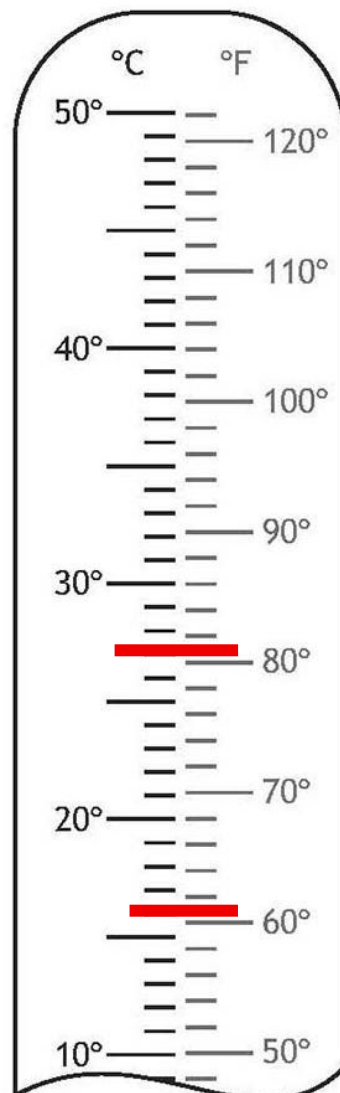
1. Hotel: $515 \times 10 = \text{£}5150 \times 0.82 = \text{£}4223$

Wi-Fi: $6 \times 10 = \text{£}60$

Breakfast: (Adults) = $25 \times 2 \times 10 = \text{£}500$ (Children) = $12.50 \times 2 \times 10 = \text{£}250$

Total: $\text{£}5033$ Yes as $\text{£}5033 < \text{£}5100$

2. **27°C**



3. 16:25 > 23:10 > 18:10

3a) Fill in the table below how many dollars they will receive.

Currency

 **USD** US dollar

Amount

£

2500

=

\$

3273

You'll pay £2500.00

Standard Rate

£1 = \$1.3092

c) $3273 - (10 \times 232.18) = \$951.20 \div 1.3092 = £726.55$